Amirhossein Shirzad

Senior Software Engineer | AI & Backend Engineer | Applied AI Researcher

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Professional Summary

Software engineer with a research background in applied AI, specializing in backend systems, retrieval-augmented generation (RAG), and production ML deployment. Experienced in designing and shipping scalable systems with Python, Django, PostgreSQL, and Docker, alongside a graduate research record in multimodal deep learning, including a peer-reviewed IEEE publication and postgraduate research training in transparent AI and NLP. Additional experience spans Android development and XR/VR systems engineering. Comfortable operating across the full lifecycle of a project, from research and architecture to production deployment.

Professional Experience

Full-stack Developer Dec 2025 – Present
Mellat Ventures

- Architected **MTForum**, an AI-powered knowledge platform (Django REST Framework, Next.js, FAISS, SBERT) implementing a RAG pipeline for semantic search and LLM-based question answering, with incremental indexing and cross-process locking for concurrent updates.
- Built and maintain **mellatventures.com**, a production VC platform (Django, PostgreSQL, Next.js), including a blog engine, newsletter system, and signal-driven cache revalidation.
- Implemented shared platform infrastructure: a secure OTP-based admin authentication system and a signed, rate-limited file-download service used across both products.
- Designed and deployed a bank branch queue-management system (FastAPI, SQLModel, Next.js) for a client demonstration.
- Own containerized deployment (Docker, Docker Compose, Nginx, Gunicorn) on Linux infrastructure, resolving networking, SSR/CSR routing, and CI/CD issues.

Co-founder & Software Engineer Jul 2023 – Nov 2025
VaraSoft

- Co-founded the company and led engineering for a multimodal AI avatar platform integrating speech recognition, text-to-speech, NLP, facial animation, and computer vision.
- Designed backend services and APIs supporting real-time, AI-driven interactive applications.
- Developed a VR banking application for Meta Quest with secure authentication and financial data visualization.
- Built cross-platform Android and Windows applications incorporating image processing and AI features.

Software Developer Jun 2017 – Aug 2020
Center for IT and Cyberspace, University of Tehran

- Led Android development for the university's official mobile application, serving 10,000+ students and staff.
- Designed backend services (Django REST Framework, PostgreSQL) for university software platforms.
- Built REST APIs, database models, and deployment automation for production systems.

Android Developer Mar 2014 – May 2017
Raadwin

- Developed and shipped multiple Android applications, including UT App, Football Sticker, YaadYaar, and Mr. Albaloo.
- Improved performance, usability, and reliability through iterative optimization and cross-functional collaboration with design and QA teams.

Software Developer Apr 2013 – Feb 2014
PulseWare

- Built an automated data-mining and monitoring system for ATM software (C#, ASP.NET).

Education

M.Sc. in Information Technology Engineering – Multimedia Systems
University of Tehran, Tehran, Iran Sep 2016 – Oct 2020

- Thesis: **Deep Learning Approach for Image and Text Sentiment Analysis** – multimodal sentiment analysis using text, images, GIFs and memes.
- GPA: 17.00/20

B.Sc. in Computer Science
University of Tehran, Tehran, Iran Sep 2008 – Jul 2013

- Thesis: **E-voting System** GPA: 14.39/20

Artificial Intelligence Research Training
University of Galway, Ireland Oct 2020 – Nov 2021

- Postgraduate research training covering Transparent AI, Visual Media Processing, NLP, Personalization and Optimization.

Publications

Shirzad, Amirhossein, Hadi Zare, and Mehdi Teimouri. "Deep Learning approach for text, image, and GIF multimodal sentiment analysis." *2020 10th International Conference on Computer and Knowledge Engineering (ICCKE), IEEE, 2020.*

Technical Skills

Programming: Python, Java, Kotlin, C#, SQL, Bash

Backend: Django, Django REST Framework, FastAPI, REST APIs, WebSockets, PostgreSQL, Redis, SQLModel

AI & Machine Learning: LLM Applications, Retrieval-Augmented Generation, Multi-Agent Systems, NLP, Computer Vision, Deep Learning, PyTorch, TensorFlow, Keras, Scikit-learn, FAISS

Frontend: Next.js, TypeScript, React, Tailwind CSS

Mobile: Android (Java/Kotlin, Jetpack Compose)

Infrastructure: Linux, Docker, Docker Compose, Nginx, Gunicorn, Git, CI/CD

XR & Multimedia: Unreal Engine, Unity, Virtual Reality, Augmented Reality, Meta Quest

Selected Projects

- **Multi-Agent Cybersecurity RAG System (CyberRAG)** – Multi-agent RAG system for cybersecurity question answering, with source code, technical report, README, and evaluation test suite; deployed to a self-managed Ubuntu VPS.
- **Bank Branch Queue Management System** – End-to-end queue-management system for bank branches, built from scratch for a client demonstration (FastAPI, SQLModel, Next.js).
- **AI Avatar Platform** – Multimodal AI avatar system integrating speech recognition, text-to-speech, facial animation, and computer vision for interactive applications.
- **Virtual Reality Banking** – Immersive VR banking application for Meta Quest with secure authentication and financial data visualization.

Awards

- **1st Place**, 3rd Fanavard Nationwide Competition in Open-source Intelligence (OSINT), Sharif University of Technology, 2016.
- **Best Student Project Award**, UT App, Second Annual Meeting for Introducing Latest IT Achievements, University of Tehran, 2017.

Languages

English: TOEFL iBT 106/120 (R:30, L:28, S:22, W:26); Duolingo English Test: 130

Persian: Native